Thesis Proposal

Title

Analysis of Using Machine Learning Application Possibilities for the Detection and Classification of Topographic Objects: A Comparative Study Using Satellite Imagery and LiDAR Data

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1. Background and Problem Statement

Bangladesh is experiencing rapid urban growth, increasing population pressure, and frequent climate-related challenges such as flooding and land-use change. These issues make accurate and up-to-date topographic information extremely important for urban planning, disaster management, infrastructure development, and environmental monitoring.

Traditionally, topographic objects like buildings, roads, vegetation, and water bodies are identified manually from satellite images, aerial photographs, or LiDAR data. However, this manual process is slow, labor-intensive, and often prone to human error—especially when dealing with large areas like Dhaka and its surrounding regions.

With the advancement of remote sensing technologies, a huge amount of data is now available from sources such as Sentinel-2 and other high-resolution imagery. Despite this, automatically extracting meaningful information from these datasets remains challenging. The complexity of Bangladesh's mixed urban and rural landscapes, frequent cloud cover, and variations in object size make accurate classification difficult.

Recent studies, including a 2026 review by Kryzia et al., show that deep learning methods such as U-Net and convolutional neural networks (CNNs) generally perform better than traditional machine learning approaches like Random Forest and Support Vector Machines (SVM). However, deep learning models require large labeled datasets and high computational power—resources that are often limited in developing countries like Bangladesh.

Because of these constraints, it is important to identify which methods are most practical and effective in the local context. This study aims to address this gap by comparing both approaches using real data from Bangladesh and providing practical recommendations.

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2. Aim and Objectives

Aim

The main aim of this research is to compare classical machine learning and deep learning techniques for automatically detecting and classifying key topographic features—specifically buildings and roads—in Bangladesh, and to develop a practical guideline for their use in real-world GIS applications.

Specific Objectives

- To review recent research on machine learning and deep learning methods for topographic mapping, including studies from 2025–2026.
- To collect and prepare remote sensing datasets for Dhaka and nearby areas using sources like Sentinel-2, Google Earth Engine, and OpenStreetMap.
- To implement classical machine learning models (e.g., Random Forest, SVM) and deep learning models (e.g., U-Net, CNN-based segmentation).
- To evaluate and compare the performance of these models using metrics such as accuracy, F1-score, and Intersection over Union (IoU).
- To analyze trade-offs between accuracy, computational cost, data requirements, and ease of implementation.
- To develop a simple decision-making framework for selecting appropriate methods under different resource conditions in Bangladesh.
- To discuss how the results can support national mapping efforts and disaster management planning.

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3. Research Questions

This study will focus on the following key questions:

1. How do classical machine learning and deep learning methods compare in detecting topographic objects in Bangladesh?

2. Which type of data (satellite imagery alone or combined with height information like LiDAR) produces better results?
3. What challenges arise when applying these techniques in Bangladesh, and how can they be addressed?
4. How can the findings be used to support automated map updating in national GIS systems?

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4. Literature Review (Brief Overview)

Previous research has shown significant progress in using machine learning and deep learning for object detection in remote sensing. The study by Kryzia et al. (2026) provides a comprehensive comparison between classical methods (such as SVM, Random Forest, and KNN) and deep learning architectures (such as CNNs, U-Net, and ResNet).

Their findings suggest that deep learning models perform better in complex environments because they can automatically learn features from data. However, these models require large training datasets and high-performance computing resources. On the other hand, classical machine learning methods are simpler, require less data, and are easier to interpret, making them suitable for smaller projects.

In the context of Bangladesh, datasets such as the Bangladesh Open Land Use/Land Cover (BOLM) dataset and the Google Open Buildings dataset provide valuable opportunities for research. However, there is still a lack of localized comparative studies and practical guidelines tailored to resource-limited environments.

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5. Proposed Methodology

Research Design

This study will follow a mixed approach, combining literature review with experimental analysis.

Data Sources

- Satellite imagery: Sentinel-2 and other available high-resolution data
- Ground truth data: OpenStreetMap, BOLM dataset, Google Open Buildings
- Additional data: LiDAR or DSM (if available from local agencies like SPARRSO or Survey of Bangladesh)

Study Area

The primary focus will be Dhaka metropolitan area and its surrounding regions.

Methods

- **Preprocessing:** Data cleaning, normalization, tiling, and augmentation using Python tools such as rasterio and geopandas
- **Classical ML:** Object-Based Image Analysis (OBIA) with Random Forest and SVM
- **Deep Learning:** Semantic segmentation using U-Net or similar architectures with TensorFlow or PyTorch
- **Evaluation:** Performance will be measured using accuracy, precision, recall, F1-score, IoU, and confusion matrices
- **Visualization:** Results will be mapped and analyzed using QGIS

Tools

Python (NumPy, scikit-learn, TensorFlow/PyTorch), QGIS, Google Earth Engine, Google Colab

Ethical Considerations

All datasets used will be open-source or publicly available. Proper attribution will be given, and no personal or sensitive data will be involved.

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6. Expected Outcomes and Significance

This research is expected to:

- Provide a clear comparison of machine learning and deep learning methods for Bangladesh
- Identify the most practical and efficient techniques for local applications
- Develop a decision-making guide for GIS professionals and government agencies
- Support automated map updating and improve planning and disaster response

The findings could also contribute to Bangladesh's National Spatial Data Infrastructure (NSDI) and may be suitable for publication in academic journals or conferences.

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7. Work Plan / Timeline

Phase	Duration
Literature review & data collection	
Model development & experiments	
Analysis & framework development	
Thesis writing	
Final submission & defense	

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8. References (Selected)

- Kryzia, K., Radziejowska, A., Adamczyk, J., & Kryzia, D. (2026). *Analysis of Machine Learning Applications for Detection and Classification of Topographic Objects*. ISPRS International Journal of Geo-Information.
- Additional references related to Sentinel-2, BOLM dataset, and Open Buildings will be included in the final thesis.